

Medicare Provider Fraud Risk Analysis | Virginia 2023

FROM: Tomorrow Lake, Data Science Analyst
RE: Specialty-Adjusted Anomaly Detection Findings - Virginia Medicare Providers, CY 2023
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Executive Summary

This analysis applies a specialty-adjusted peer-comparison method to all Medicare Part B providers in Virginia for calendar year 2023. The methodology screens for billing patterns that deviate substantially from specialty norms across four dimensions: payment per service, services per beneficiary, payment per beneficiary, and submission-to-allowed ratio. **Of 25,239 providers analyzed, 155 (0.6%) were flagged on three or more dimensions, representing \$108 million in Medicare payments and a high-priority audit population.** An additional 2,208 providers (8.7%) flagged on two dimensions warrant secondary review.

The methodology is interpretable, peer-aware, and aligned with NIST AI Risk Management Framework guidance on explainable detection systems. Every flagged provider can be characterized in plain language ("this provider bills X times the specialty median for Y"), supporting due-process requirements in any subsequent enforcement action.

Key Findings

- **Payment concentration is extreme.** The top 1% of providers (317) capture 28.6% of all Virginia Medicare payments. The top 5% capture 48.7%. Audit prioritization focused on this small population offers substantial leverage on total program spend.
- **Specialty-adjusted scoring eliminates the false-positive problem.** An unadjusted screen flags oncologists, rheumatologists, and infusion-heavy specialties as outliers because they legitimately bill at high intensity. The peer-comparison method removes this noise and surfaces a different, more diverse set of high-risk providers across 15+ specialties.
- **Risk concentrates in identifiable specialty clusters.** Flagged providers concentrate in Diagnostic Radiology, Ophthalmology, Radiation Oncology, Ambulance Services, and Clinical Laboratory. These five specialties account for the majority of High and Critical risk flags and represent natural starting points for focused audit work.
- **Geographic patterns mirror payment volume.** Risk is highest in Richmond, Northern Virginia, Hampton Roads, and Fredericksburg - the same regions that account for approximately 42% of Virginia Medicare payments overall. This is expected and confirms that audit resources should follow payment density, not be uniformly distributed.

Risk Distribution

Risk Category	Providers	Medicare Payments
Critical (4 flags)	9	\$5.3M
High (3 flags)	146	\$102.5M
Elevated (2 flags)	2,208	\$770.8M
Moderate (1 flag)	3,724	\$528.7M
Low (0 flags)	19,152	\$1.50B

Recommendations

- **Initiate review of the 155 High and Critical risk providers.** Prioritize the top 15 by total payments, who collectively represent \$53.6 million in Medicare disbursements. Provider-level findings are documented in the supporting dashboard and dataset.
- **Establish specialty-specific thresholds for ongoing screening.** Specialty norms differ by orders of magnitude. A single statewide threshold either misses real fraud in low-billing specialties or generates excessive false positives in high-billing ones.
- **Pair detection with monitoring infrastructure.** Fraud schemes evolve. Static models trained on historical data degrade as adversaries adapt. Continuous retraining and out-of-time validation should be treated as core operational requirements, not optional refinements.
- **Treat threshold selection as a governance decision.** The 2x specialty-median threshold used here is defensible but not unique. Tightening it to 3x reduces investigator workload but lets more fraud through; loosening it captures more fraud but increases false positives. This trade-off should be made explicit and revisited annually.

Methodology Notes

Source: CMS "Medicare Physician & Other Practitioners - by Provider" public-use file, calendar year 2023. Filtered to Virginia (state code VA), 31,662 providers. Analysis-ready population of 25,239 after applying minimum thresholds (50 or more beneficiaries per provider, 30 or more peers per specialty).

Method: For each of four billing metrics, compute the specialty-level median; calculate each provider's deviation ratio (provider value divided by specialty median); flag providers exceeding 2x the median; aggregate flag count to a 0-4 risk score.

Limitations: This is an unsupervised screening tool, not a determination of fraud. Flagged providers warrant further investigation, not adverse action. Some specialties (e.g., Clinical Laboratory) operate under business models that produce statistically unusual patterns by design and should be evaluated separately. The 50-beneficiary minimum reduces but does not eliminate small-sample sensitivity.

Supporting Materials

- Interactive dashboard:
public.tableau.com/app/profile/tomorrow.lake5414/viz/MedicareProviderFraudRiskAnalysis-Virginia2023
- Code, data, and queries: github.com/tomorrowlake3/Medicare_Provider_Fraud_Analysis
- Companion paper: Anomaly Detection at Scale: Methods and Trade-offs for Identifying Improper Payments in Federal Programs